

Prismoidal Volumes and Homothetic Corrections for Similar Irregular Polygonal End Areas

Theory, Surveying Practice, Numerical Integration Context, and Application to Irregular Parallel Polygons

Abstract

Prismoidal volume computation is a central technique in surveying, earthworks, mining, reservoir estimation, and solid geometry. The classical prismoidal formula uses the two end areas of a solid, the area of the middle section, and the perpendicular spacing between the end planes. Its practical difficulty is not the formula itself, but the reliable determination of the middle area when the end areas are irregular, when field measurements are sparse, or when the intermediate section is physically hard to construct. This article develops a detailed treatment of prismoidal volumes with special attention to the case in which the two end faces are irregular polygons of the same plane shape, lie in parallel planes, and differ only by a uniform change of scale. In that case the missing middle area is not the arithmetic mean of the end areas. It is the area associated with the mean of the linear scales, namely

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2.$$

Substitution in the prismoidal formula gives the homothetic-frustum volume

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2),$$

or, equivalently, the correction to subtract from the average end-area volume is

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

The article also explains why this is not the same as the usual three-level road-section correction, why no unique correction can be obtained from only A_1 and A_2 for arbitrary non-similar irregular polygons, and when triangulation, finite-element summation, or numerical integration should be preferred. The discussion synthesises the supplied material on the prismoidal formula, end-area method, prismoidal correction, frusta, prismatoids, numerical integration, and historical solid-volume formulae (Deakin and Hunter 2009; Venkatramaiah 2011; Agor 2015; Earthworks Module n.d.; Volume Calculations Manual 2024; Malvic et al. 2014; Olu 2020; Heydari and Muroi 2023; Wentworth n.d.).

Keywords: prismoidal formula; prismatoid; frustum; homothety; irregular polygon; end-area method; middle area; Simpson's rule; earthworks; numerical integration.

Introduction

Volume computation from cross-sections is one of the oldest and most persistent geometric problems in civil engineering, surveying, mining, geology, and hydrology. In earthworks, dredging, embankment construction, cut-and-fill balancing, reservoir capacity estimation, and geological volume assessment, the available data are often not a complete three-dimensional surface model. Instead, the data are a sequence of measured or designed cross-sectional areas. The engineering problem is then to infer a volume between sections from partial geometric information.

The two most common formulae in this context are the average end-area formula and the prismoidal formula. The average end-area formula is simple and requires only the two end areas; the prismoidal formula is more exact for prismoids but requires the area of a middle section. Surveying references repeatedly emphasise this operational distinction: the end-area method is convenient, while the prismoidal method is more exact when the assumed prismoid geometry is satisfied and when the middle section can be determined (Deakin and Hunter 2009; Agor 2015; Venkatramaiah 2011; Volume Calculations Manual 2024).

This article addresses a specific practical problem: the two end areas A_1 and A_2 are irregular polygons, but they have the same plane shape, lie in parallel planes, and differ only by uniform linear scale. Their levels may be slightly different; therefore the relevant distance between them is the perpendicular distance between their planes. The physical middle polygon is difficult to construct or measure. The question is: which correction should be applied to the prismoidal formula, or to the average end-area result, without explicitly drawing the middle section?

The answer is a homothetic-frustum correction. Because the end polygons have the same shape, the linear dimensions change linearly between corresponding vertices; because area is quadratic in linear scale, the middle area is obtained from the average of the square roots of the end areas, not from their arithmetic mean. This is the same mathematical structure as the frustum of a pyramid and the conical frustum, generalised to any similar irregular polygon.

The conclusion is therefore operationally simple but geometrically strict:

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2$$

and

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

If the calculation begins with the average end-area volume,

$$V_{EA} = \frac{L}{2} (A_1 + A_2),$$

then the corresponding positive correction to subtract is

$$C = V_{EA} - V = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

The correction is exact under the stated homothetic assumptions. It is not a general correction for arbitrary irregular polygons. If the polygons differ in shape, twist, local distortion, number of vertices, vertex order, or non-uniform scaling, the two end areas alone are insufficient to determine the missing middle area.

Homothetic irregular polygonal prismoid/frustum

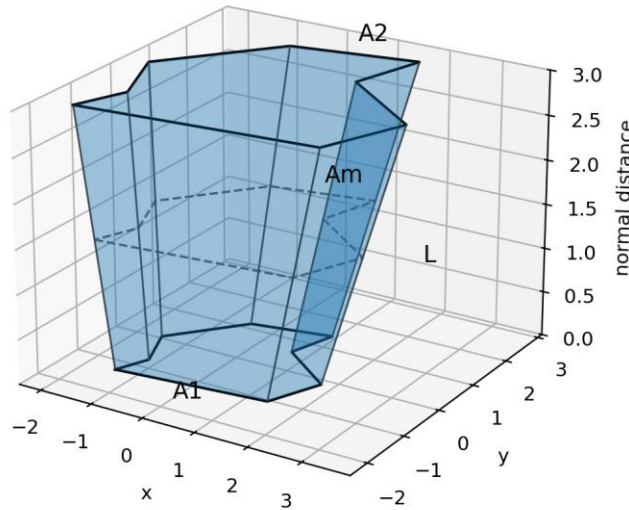


Figure 1. Homothetic irregular polygonal prismoid/frustum: two similar irregular end polygons in parallel planes, with a middle section at the average linear scale.

Terminology and geometric setting

Polyhedra, prisms, prismatoids, prismoids, and frusta

A polyhedron is a solid bounded by plane faces and straight edges. In elementary geometry, a prism is a polyhedron with two parallel congruent polygonal bases and lateral parallelogram faces; its volume is the base area multiplied by the perpendicular height (Volume Calculations Manual 2024; Bonjorno, Giovanni Junior, and Camara de Sousa 2020). If B is the base area and h is the perpendicular distance between bases, then

$$V = Bh.$$

A pyramid is a polyhedron with a polygonal base and triangular lateral faces meeting at an apex. Its volume is

$$V = \frac{1}{3}Bh.$$

A frustum of a pyramid or cone is the portion between the base and a plane section parallel to the base, after the apex has been removed. Educational modules on frusta and prismatoids define a frustum as the portion of a cone or pyramid lying between parallel cutting planes (Frustum and Prismatoid Module n.d.; Volumes of Solids Teaching Slides n.d.). For a frustum with parallel similar bases of areas A_1 and A_2 and perpendicular height L , the standard area-based formula is

$$V = \frac{L}{3}(A_1 + \sqrt{A_1 A_2} + A_2).$$

A prismatoid is usually defined as a polyhedron whose vertices lie in two parallel planes. A prismoid, as used in engineering surveying, is a solid bounded by two parallel end faces and plane side faces joining them. The end faces need not be similar and need not have the same number of sides (Deakin and Hunter 2009; Agor 2015; Wentworth n.d.). Deakin and Hunter describe the assumed earthwork solid between cross-sections as a prismoid with parallel plane end-faces and side faces extending over the full length, and they note that prismoids may be decomposed into prisms, wedges, and pyramids (Deakin and Hunter 2009). Agor similarly states that a prismoid has two parallel end faces and longitudinal plane faces (Agor 2015).

The prismoidal formula is

$$V_P = \frac{L}{6}(A_1 + 4A_m + A_2),$$

where A_1 and A_2 are the end areas, A_m is the area of the middle section parallel to the ends, and L is the perpendicular distance between the end planes.

Wentworth's scanned article on graphic computation of contoured solids gives the same formula in the notation A , M , B and h :

$$V = \frac{h}{6}(A + 4M + B),$$

where A and B are the end areas and M is the middle area; that article is especially relevant here because it explicitly treats bases that may be of any shape and need not be similar (Wentworth n.d.).

The meaning of distance between slightly different levels

For parallel horizontal sections at elevations z_1 and z_2 , the distance between planes is simply

$$L = |z_2 - z_1|.$$

For two inclined but parallel planes written as

$$z = ax + by + c_1$$

and

$$z = ax + by + c_2,$$

the perpendicular spacing is

$$L = \frac{|c_2 - c_1|}{\sqrt{a^2 + b^2 + 1}}.$$

More generally, if the two parallel planes are written in implicit form as

$$\mathbf{n} \cdot \mathbf{x} = d_1$$

and

$$\mathbf{n} \cdot \mathbf{x} = d_2,$$

then the perpendicular distance is

$$L = \frac{|d_2 - d_1|}{\|\mathbf{n}\|}.$$

This distinction matters. If the volume is intended as a true geometric volume between two parallel planes, L is the normal distance. If a project intentionally uses vertical projection, then the vertical difference may be used, but the computed quantity is not the same as the true normal volume.

Classical volume formulae

Prism, wedge, pyramid, and frustum

The elementary formulae can be interpreted as special cases of integration of area along a height direction. If the cross-sectional area is constant, the solid is a prism and

$$V = Bh.$$

If the area varies linearly with distance, the trapezoidal rule is exact:

$$V = \frac{h}{2}(A_1 + A_2).$$

If the area varies quadratically with distance, Simpson's rule is exact:

$$V = \frac{h}{6}(A_1 + 4A_m + A_2).$$

A pyramid is a limiting case in which one end area is zero. Setting $A_2 = 0$ in the frustum formula gives

$$V = \frac{h}{3}A_1.$$

A frustum of a pyramid or cone with similar parallel bases has the area-based expression

$$V = \frac{h}{3}(A_1 + \sqrt{A_1A_2} + A_2).$$

Olu reviews the Egyptian-style expression for a pyramidal frustum in terms of the two end areas and the square root of their product, and develops diagonal-based expressions for square and rectangular frusta under specified geometric assumptions (Olu 2020). For a square frustum using base diagonals D and d , the formula reported is

$$V_{SF} = \frac{h}{6}(D^2 + Dd + d^2).$$

For the special rectangular-frustum model considered by Olu, with rectangular sides in ratio 2: 1, the diagonal-based expression is

$$V_{RF} = \frac{2h}{15}(D^2 + Dd + d^2).$$

These diagonal forms are special parameterisations of similar-base frusta. The general engineering expression remains the area formula with A_1 , A_2 , and $\sqrt{A_1A_2}$.

Average end-area formula

The average end-area formula, also called the trapezoidal formula in surveying texts, is

$$V_{EA} = \frac{L}{2}(A_1 + A_2).$$

For a sequence of n equally spaced sections with spacing D , the composite average end-area expression is

$$V_{EA} = \frac{D}{2}(A_1 + 2A_2 + 2A_3 + \cdots + 2A_{n-1} + A_n).$$

Equivalently,

$$V_{EA} = D \left(\frac{A_1 + A_n}{2} + \sum_{i=2}^{n-1} A_i \right).$$

Agor presents the end-area formula as the product of length by the average of the end areas, and Venkatramaiah notes that it is convenient but exact only under specific geometric conditions, such as prismoids composed of prisms and wedges (Agor 2015; Venkatramaiah 2011). Deakin and Hunter show that the end-area formula can be exact,

under-estimate, or over-estimate, depending on the prismoid composition and, particularly, the presence and orientation of pyramidal components (Deakin and Hunter 2009).

Prismoidal formula and Simpson's rule

The prismoidal formula is

$$V_P = \frac{L}{6}(A_1 + 4A_m + A_2).$$

For a sequence of equally spaced sections, the composite prismoidal formula is Simpson's rule. With an odd number of cross-sections, equal spacing D , and areas $A_1, A_2, \dots, A_n$, it is

$$V_P = \frac{D}{3} \left(A_1 + A_n + 4 \sum_{i=2,4,6,\dots}^{n-1} A_i + 2 \sum_{i=3,5,7,\dots}^{n-2} A_i \right).$$

Equivalently, written in the common surveying form,

$$V_P = \frac{D}{3} [A_1 + 4(A_2 + A_4 + \dots) + 2(A_3 + A_5 + \dots) + A_n].$$

Agor explicitly identifies the prismoidal formula for a series of cross-sections as Simpson's rule for volumes and notes that an odd number of sections is required for its direct composite application (Agor 2015). Venkatramaiah similarly describes the use of alternate sections as mid-sections when equally spaced data are available (Venkatramaiah 2011).

General prismoidal correction

Correction as the difference between end-area and prismoidal volume

Let the positive correction to subtract from the average end-area volume be

$$C = V_{EA} - V_P.$$

Substitution of the two formulae gives

$$C = \frac{L}{2}(A_1 + A_2) - \frac{L}{6}(A_1 + 4A_m + A_2).$$

After collecting terms,

$$C = \frac{L}{6}(3A_1 + 3A_2 - A_1 - 4A_m - A_2).$$

Therefore,

$$C = \frac{L}{6}(2A_1 + 2A_2 - 4A_m),$$

and hence

$$C = \frac{L}{3}(A_1 + A_2 - 2A_m).$$

With this sign convention,

$$V_P = V_{EA} - C.$$

The uploaded earthworks module derives the same correction expression, beginning with $V_c = V_E - V_p$ and reducing it to $L(A_1 + A_2 - 2A_m)/3$ (Earthworks Module n.d.). Agor presents the prismoidal correction as the difference between the end-area and prismoidal volumes (Agor 2015). Deakin and Hunter define an algebraic correction as $PC = V_P - V_{EA}$; this is the same quantity with opposite sign (Deakin and Hunter 2009). Under Deakin and Hunter's algebraic convention,

$$PC = V_P - V_{EA} = -\frac{L}{3}(A_1 + A_2 - 2A_m).$$

This sign convention must be made explicit in reports. In this article, the symbol C denotes the positive amount to subtract from the average end-area result when the homothetic-frustum case gives a smaller volume.

Why the correction is not generally known from A_1 and A_2

The formula

$$C = \frac{L}{3}(A_1 + A_2 - 2A_m)$$

shows that the correction depends on A_m . Therefore no correction can be uniquely determined from A_1 and A_2 alone unless an additional geometric model fixes the middle area. This is the central point in practical use.

The average end-area formula implicitly assumes

$$A_m = \frac{A_1 + A_2}{2}.$$

If this is true, then

$$C = \frac{L}{3}\left(A_1 + A_2 - 2 \cdot \frac{A_1 + A_2}{2}\right) = 0,$$

and

$$V_P = V_{EA}.$$

However, in a pyramidal or homothetic frustum, area varies quadratically with the linear scale, so A_m is generally less than the arithmetic mean of A_1 and A_2 when the two end areas differ. Venkatramaiah explicitly notes that the middle area can be found by averaging

corresponding linear dimensions, not by averaging the end areas, and that for simple geometrical figures the average linear dimensions represent the mid-section dimensions (Venkatramaiah 2011). Appendix F similarly states that the midpoint area is determined from averaging corresponding linear heights and widths, not from averaging the areas (Volume Calculations Manual 2024).

Geometry-specific corrections and their limitations

Some surveying and earthworks texts give simplified corrections for particular road cross-section types. For example, the earthworks module states a three-level-section correction in the form

$$V_{cp} = \frac{L}{12}(C_1 - C_2)(D_1 - D_2),$$

where the symbols refer to particular section heights and widths in the three-level road-section geometry (Earthworks Module n.d.). For a level section of constant formation width b and side slope s : 1, if the section area is

$$A = h(b + sh),$$

then, using $h_m = (h_1 + h_2)/2$, the positive correction becomes

$$C = \frac{Ls}{6}(h_1 - h_2)^2.$$

These formulae are useful only because the geometry defines how A_m is obtained. Deakin and Hunter warn that practical prismoidal corrections based only on end-section data are correct only for certain solids and can be incorrect when the actual section differs from the assumed section type (Deakin and Hunter 2009). Therefore a three-level road-section correction should not be used for arbitrary irregular polygons unless the irregular polygons have first been shown to correspond to that particular section model.

Homothetic irregular polygons

Definition of the case studied

The central case considered in this article is the following:

1. The two end faces lie in parallel planes.
2. Their areas are A_1 and A_2 .
3. Each end face is an irregular polygon.
4. The two polygons have the same internal shape and the same vertex order.
5. The second polygon is obtained from the first by a uniform linear scale factor, with possible translation in the parallel plane.
6. Corresponding vertices are connected by straight generatrices.
7. The distance L is the perpendicular distance between the two planes.

This is a homothetic frustum. It is not necessary for the polygon to be regular; it may be any irregular polygon. What matters is that the transformation from one end polygon to the other is a uniform scaling of all linear dimensions.

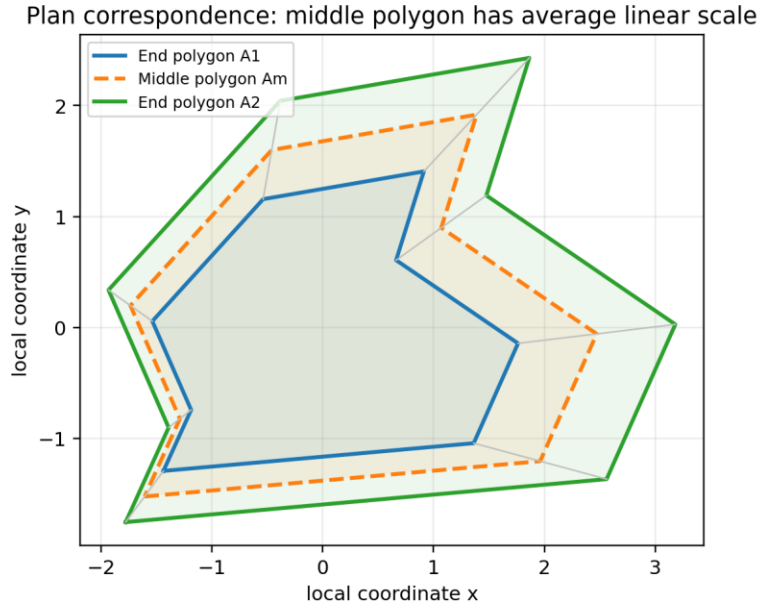


Figure 2. Plan correspondence between the lower end polygon, middle polygon, and upper end polygon. The middle polygon is obtained by the mean linear scale, not by the mean area.

Area of a uniformly scaled polygon

Let P be a reference polygon of area K . If all its linear dimensions are multiplied by a scale factor λ , its area becomes

$$A(\lambda) = K\lambda^2.$$

This is true for any plane polygon, including irregular polygons. It follows from the fact that both coordinate directions scale by λ , so the determinant of the planar scaling transformation is λ^2 .

For a polygon with vertices (x_i, y_i) , $i = 1, 2, \dots, n$, the signed area is obtained by the shoelace formula

$$A = \frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right|,$$

with the convention

$$(x_{n+1}, y_{n+1}) = (x_1, y_1).$$

If the coordinates are scaled as

$$(x_i, y_i) \mapsto (\lambda x_i, \lambda y_i),$$

then each term in the shoelace sum is multiplied by λ^2 :

$$(\lambda x_i)(\lambda y_{i+1}) - (\lambda x_{i+1})(\lambda y_i) = \lambda^2(x_i y_{i+1} - x_{i+1} y_i).$$

Therefore

$$A(\lambda) = \lambda^2 A(1).$$

This derivation proves that the correction is independent of whether the polygon is regular or irregular. Only homothety matters.

Linear scale interpolation between end sections

Let the lower end polygon have scale λ_1 and area

$$A_1 = K\lambda_1^2.$$

Let the upper end polygon have scale λ_2 and area

$$A_2 = K\lambda_2^2.$$

At a normal coordinate x measured from the first plane, define

$$u = \frac{x}{L}.$$

If corresponding vertices are joined by straight segments, the scale varies linearly:

$$\lambda(v) = (1 - v)\lambda_1 + v\lambda_2.$$

The area at that intermediate section is therefore

$$A(v) = K[(1 - v)\lambda_1 + v\lambda_2]^2.$$

At the mid-section, $v = 1/2$:

$$\lambda_m = \frac{\lambda_1 + \lambda_2}{2}.$$

Thus

$$A_m = K\left(\frac{\lambda_1 + \lambda_2}{2}\right)^2.$$

Because

$$\sqrt{A_1} = \sqrt{K}\lambda_1$$

and

$$\sqrt{A_2} = \sqrt{K}\lambda_2,$$

the middle area can be written only in terms of A_1 and A_2 :

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2.$$

Expanding this expression,

$$A_m = \frac{A_1 + 2\sqrt{A_1 A_2} + A_2}{4}.$$

Therefore

$$4A_m = A_1 + 2\sqrt{A_1 A_2} + A_2.$$

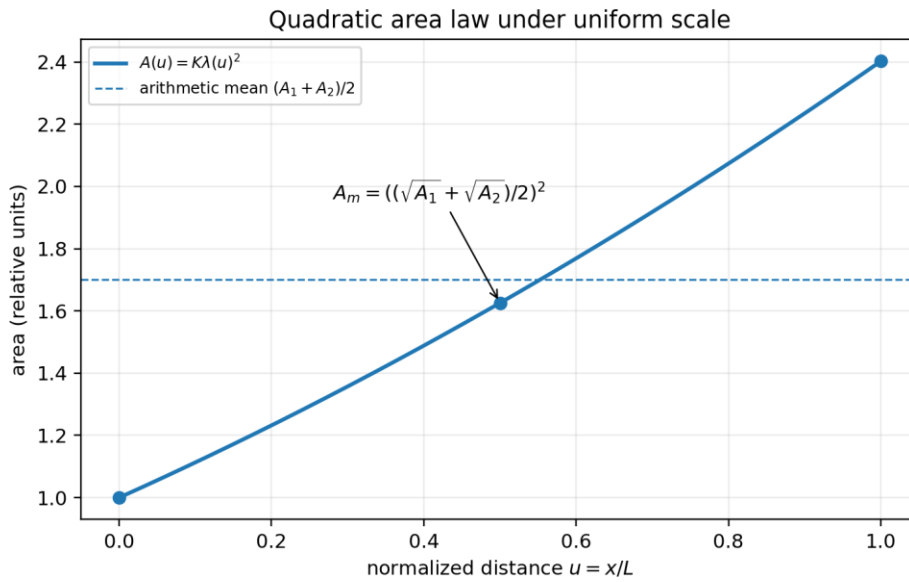


Figure 3. Area varies quadratically with linear scale. The mid-section area is below the arithmetic mean of the end areas when the end areas are unequal.

Substitution into the prismoidal formula

The prismoidal formula is

$$V_P = \frac{L}{6}(A_1 + 4A_m + A_2).$$

Substituting

$$4A_m = A_1 + 2\sqrt{A_1 A_2} + A_2$$

gives

$$V_P = \frac{L}{6}(A_1 + A_1 + 2\sqrt{A_1 A_2} + A_2 + A_2).$$

Therefore

$$V_P = \frac{L}{6} (2A_1 + 2\sqrt{A_1 A_2} + 2A_2),$$

or

$$\boxed{V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2)}.$$

This is the same as the pyramid-frustum formula. Deakin and Hunter derive the pyramid-frustum formula from similar triangles and areas and state that, although they illustrate it using a triangular pyramid, it applies to frustums of other pyramids because they can be decomposed into triangular pyramid frustums (Deakin and Hunter 2009). The argument here extends the same principle to any irregular polygon whose end faces are similar and homothetic.

Derivation of the correction to the average end-area formula

The average end-area formula is

$$V_{EA} = \frac{L}{2} (A_1 + A_2).$$

The homothetic-frustum volume is

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

The correction to subtract is

$$C = V_{EA} - V.$$

Substitution gives

$$C = \frac{L}{2} (A_1 + A_2) - \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

With common denominator 6,

$$C = \frac{L}{6} [3(A_1 + A_2) - 2(A_1 + \sqrt{A_1 A_2} + A_2)].$$

Thus

$$C = \frac{L}{6} (A_1 + A_2 - 2\sqrt{A_1 A_2}).$$

Since

$$A_1 + A_2 - 2\sqrt{A_1 A_2} = (\sqrt{A_2} - \sqrt{A_1})^2,$$

the correction is

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

Therefore

$$V = V_{EA} - \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

The correction is zero only if $A_1 = A_2$:

$$A_1 = A_2 \Rightarrow C = 0.$$

If $A_1 \neq A_2$, then

$$C > 0,$$

so the average end-area method overestimates the homothetic-frustum volume.

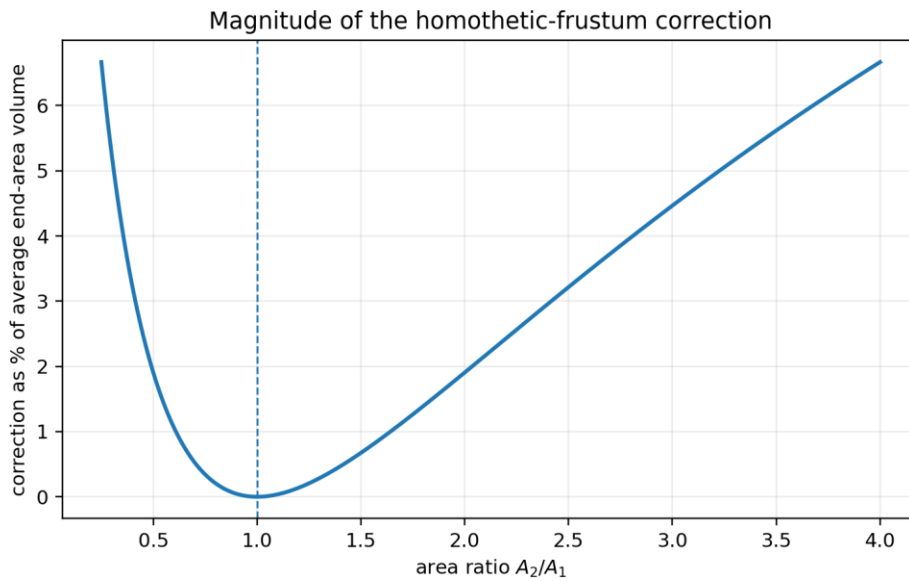


Figure 4. Relative magnitude of the correction as a function of the area ratio A_2/A_1 for $L = 1$ and $A_1 = 1$.

Exact integral derivation

Area function for a homothetic polygonal frustum

The homothetic volume may also be obtained by direct integration. With $v = x/L$, the area function is

$$A(v) = K[(1 - v)\lambda_1 + v\lambda_2]^2.$$

Let

$$\Delta\lambda = \lambda_2 - \lambda_1.$$

Then

$$\lambda(v) = \lambda_1 + v\Delta\lambda,$$

and

$$A(v) = K(\lambda_1 + v\Delta\lambda)^2.$$

Expanding,

$$A(v) = K(\lambda_1^2 + 2\lambda_1\Delta\lambda v + \Delta\lambda^2 v^2).$$

The volume is

$$V = L \int_0^1 A(v) dv.$$

Substitution gives

$$V = LK \int_0^1 (\lambda_1^2 + 2\lambda_1\Delta\lambda v + \Delta\lambda^2 v^2) dv.$$

Thus

$$V = LK \left(\lambda_1^2 + \lambda_1\Delta\lambda + \frac{\Delta\lambda^2}{3} \right).$$

Because $\lambda_2 = \lambda_1 + \Delta\lambda$, this expression can be rearranged as

$$V = \frac{LK}{3} (\lambda_1^2 + \lambda_1\lambda_2 + \lambda_2^2).$$

Using

$$A_1 = K\lambda_1^2,$$

$$A_2 = K\lambda_2^2,$$

and

$$\sqrt{A_1 A_2} = K\lambda_1\lambda_2,$$

we obtain

$$\boxed{V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2)}.$$

Why Simpson's rule is exact here

Simpson's rule is exact for any quadratic area function. The area function for a homothetic polygonal frustum is quadratic in v , as shown above. Therefore the prismoidal formula, which is Simpson's one-panel rule for volume, gives the exact integral:

$$\int_0^L A(x) dx = \frac{L}{6} \left[A(0) + 4A\left(\frac{L}{2}\right) + A(L) \right].$$

In prismoidal notation,

$$A(0) = A_1,$$

$$A\left(\frac{L}{2}\right) = A_m,$$

and

$$A(L) = A_2.$$

Hence

$$V = \frac{L}{6} (A_1 + 4A_m + A_2).$$

This explains why the prismoidal formula and the homothetic-frustum formula agree exactly in this case.

Numerical example

Consider two irregular but homothetic polygonal end sections in parallel horizontal planes. Suppose

$$A_1 = 100.00 \text{ m}^2,$$

$$A_2 = 225.00 \text{ m}^2,$$

and

$$L = 10.00 \text{ m}.$$

The square-root areas are

$$\sqrt{A_1} = 10.00 \text{ m},$$

and

$$\sqrt{A_2} = 15.00 \text{ m}.$$

The middle area is

$$A_m = \left(\frac{10.00 + 15.00}{2} \right)^2.$$

Therefore

$$A_m = 12.50^2 = 156.25 \text{ m}^2.$$

The arithmetic mean of the end areas would be

$$\frac{A_1 + A_2}{2} = \frac{100.00 + 225.00}{2} = 162.50 \text{ m}^2.$$

This is larger than the correct homothetic middle area. The prismoidal volume is

$$V = \frac{10.00}{6} (100.00 + 4 \times 156.25 + 225.00).$$

Thus

$$V = \frac{10.00}{6} (950.00) = 1583.33 \text{ m}^3.$$

Using the homothetic-frustum expression directly,

$$V = \frac{10.00}{3} (100.00 + \sqrt{100.00 \times 225.00} + 225.00),$$

which gives

$$V = \frac{10.00}{3} (100.00 + 150.00 + 225.00) = 1583.33 \text{ m}^3.$$

The average end-area volume is

$$V_{EA} = \frac{10.00}{2} (100.00 + 225.00) = 1625.00 \text{ m}^3.$$

The correction to subtract is

$$C = \frac{10.00}{6} (15.00 - 10.00)^2.$$

Therefore

$$C = \frac{10.00}{6} \times 25.00 = 41.67 \text{ m}^3.$$

Finally,

$$V = V_{EA} - C = 1625.00 - 41.67 = 1583.33 \text{ m}^3.$$

This example demonstrates the practical consequence: using the arithmetic mean area as the middle area would eliminate the correction and would return the average end-area volume, which is not correct for homothetic scaling.

Relationship with surveying practice

Why the method is compatible with conventional prismoidal theory

The homothetic correction does not replace the prismoidal formula. It supplies the missing A_m when the mid-section is hard to draw but the end polygons are known to be scaled versions of the same shape. It is therefore an application of the standard formula rather than a separate empirical method.

Surveying texts often say that the mid-section dimensions can be obtained by averaging corresponding linear dimensions of the end sections (Venkatramaiah 2011; Volume Calculations Manual 2024). The homothetic irregular polygon case is the most general form of that statement for similar irregular polygons: every corresponding dimension is governed by the same linear scale, so the middle area follows from the mean scale. The result is exact under straight generatrices.

Equivalent regular-section substitution

Venkatramaiah discusses a practical treatment of irregular sections by substituting equivalent regular sections of the same area, then averaging dimensions to obtain a middle area (Venkatramaiah 2011). That approach is a practical engineering approximation. It may be acceptable when the irregular shape is not exactly known or when equivalent level, two-level, or three-level sections are being used to fit road-earthwork conventions.

However, equivalent-area substitution is not a geometric identity. Two plane figures can have the same area and very different dimensions, centroids, moments, boundary correspondence, and side-face structure. Figure 5 illustrates this distinction. A square of the same area as an irregular polygon is area-equivalent, but it is not shape-equivalent.

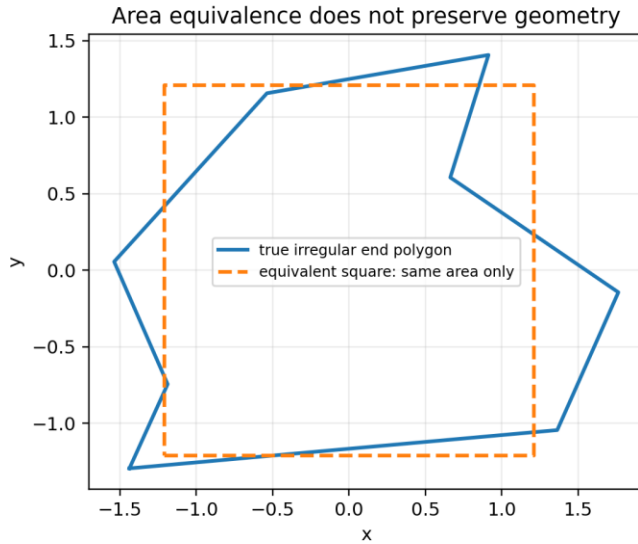


Figure 5. Area equivalence does not preserve geometry. An equivalent square can have the same area as an irregular polygon without reproducing the corresponding dimensions or side-face geometry.

For the homothetic case, no equivalent regular section is required. The square-root area relationship already recovers the correct middle area from the true irregular areas because the shapes are similar.

Why the usual road-section correction is not transferable

The three-level road-section correction is derived for a very specific geometry. It assumes the form of the road section, the formation width, side slopes, transverse ground slopes, and the relationships between heights and widths. Its correction formula is not a general prismoidal correction for all irregular polygons. Deakin and Hunter's warning is directly relevant: a prismoidal correction is correct only when the actual solid belongs to the geometric class for which that correction was derived (Deakin and Hunter 2009).

Therefore, for irregular but homothetic polygons, the correct procedure is not

$$V_{cp} = \frac{L}{12} (C_1 - C_2)(D_1 - D_2),$$

unless C_1 , C_2 , D_1 , and D_2 actually represent a valid three-level road-section model. The correct homothetic correction is

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

Arbitrary irregular polygons: why there is no end-area-only correction

Non-uniqueness of the middle area

If the two end polygons are not homothetic, the values A_1 and A_2 do not determine A_m . Many solids can have the same end areas but different middle areas and different volumes. For example, a solid may bulge in the middle, pinch in the middle, twist, change the vertex correspondence, or transition through a different intermediate geometry while retaining the same end areas.

Mathematically, the prismoidal correction requires

$$C = \frac{L}{3}(A_1 + A_2 - 2A_m).$$

If A_m is unknown and not constrained by a geometric model, C is unknown. Any formula using only A_1 and A_2 necessarily imports an additional assumption about the area function between the sections.

Possible methods when homothety is absent

When the end polygons are not the same plane shape, or when the correspondence between vertices is not uniform, the recommended methods are:

1. Construct or survey the actual mid-section and apply the prismoidal formula.
2. Split the polygons into corresponding triangles or tetrahedral/prismatic finite elements.
3. Build a triangulated irregular network (TIN) between the sections and sum element volumes.
4. Insert additional intermediate sections and integrate the area function numerically.
5. Use the average end-area method only as an approximation and state the assumption.

Deakin and Hunter describe finite-element volume summation as a method designed to overcome limitations of end-area volumes, with the finite-element volume written as a sum of small elements (Deakin and Hunter 2009):

$$V_{FE} = \sum_{j=1}^n A_j \delta x.$$

In the limit,

$$V = \lim_{\delta x \rightarrow 0} \sum_{j=1}^n A_j \delta x = \int_0^L A(x) dx.$$

This integral perspective is also central to numerical volume calculation in geological structures. Malvic et al. state that volumes are often calculated by integrating area

functions with respect to distance, and they apply trapezoidal and Simpson rules to irregular anticline and reservoir-volume contexts (Malvic et al. 2014).

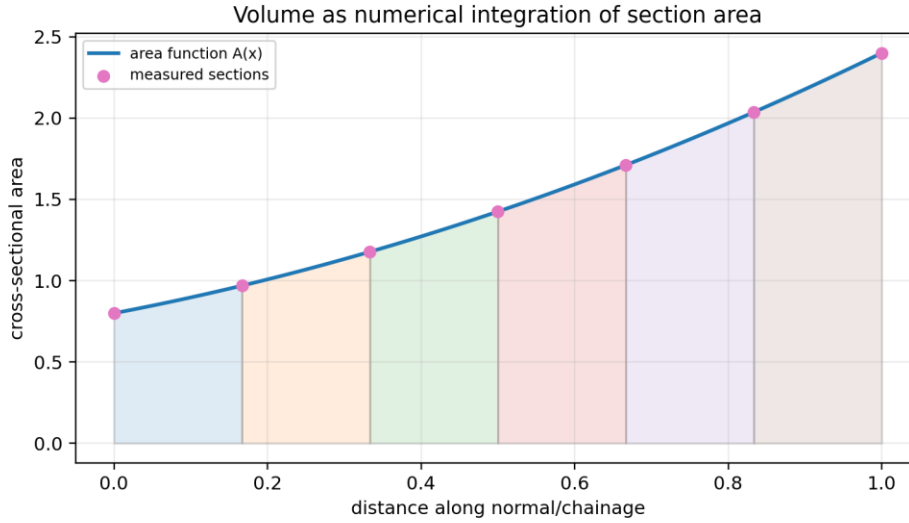


Figure 6. Volume as numerical integration of a measured or modelled area function. The trapezoidal and Simpson rules approximate the integral from discrete section areas.

Numerical integration context

Definite integral form of volume

The most general section-based expression for volume is

$$V = \int_{x_0}^{x_n} A(x) dx,$$

where $A(x)$ is the area of the section perpendicular to the selected axis or normal direction. If exact integration is possible, it should be used. In practice, $A(x)$ is often known only at discrete stations. Then numerical integration is required (Malvic et al. 2014).

Trapezoidal rule

For equally spaced stations with spacing d and areas $A_0, A_1, \dots, A_n$, the trapezoidal rule is

$$V_T = d \left[\frac{A_0 + A_n}{2} + \sum_{i=1}^{n-1} A_i \right].$$

Equivalently,

$$V_T = \frac{d}{2} (A_0 + 2A_1 + 2A_2 + \dots + 2A_{n-1} + A_n).$$

For a single interval, this reduces to

$$V_T = \frac{L}{2}(A_1 + A_2).$$

Simpson's one-third rule

For an even number of intervals and equal spacing d , Simpson's one-third rule is

$$V_S = \frac{d}{3} \left[A_0 + A_n + 4 \sum_{i=1,3,5,\dots}^{n-1} A_i + 2 \sum_{i=2,4,6,\dots}^{n-2} A_i \right].$$

For one prismoidal interval of length $L = 2d$, this is

$$V_S = \frac{L}{6}(A_1 + 4A_m + A_2).$$

Thus the prismoidal formula is Simpson's rule applied to the area function over a single span.

Simpson's three-eighths rule

When three equal subintervals are used, Simpson's three-eighths rule is

$$V_{3/8} = \frac{3d}{8}(A_0 + 3A_1 + 3A_2 + A_3).$$

This formula is less central in ordinary two-end-section prismoidal work, but it belongs to the same family of Newton-Cotes integration methods.

Accuracy and data quality

Numerical integration does not overcome inaccurate geometry. Malvic et al. note that practical data have limited accuracy, so the accuracy of the result cannot exceed the accuracy of the input data (Malvic et al. 2014). In earthworks, the same principle applies: a higher-order formula cannot compensate for poorly located sections, incorrect polygon areas, inconsistent datums, or invalid assumptions about intermediate geometry.

Historical and mathematical background

The supplied sources show that prismoidal and frustum formulae are not modern inventions. Deakin and Hunter discuss the historical presence of the prismoidal formula in ancient Egyptian mathematics, including the Moscow Mathematical Papyrus, and present a Newtonian proof using a point in the middle plane to decompose the prismoid into pyramids (Deakin and Hunter 2009). Olu also refers to the Egyptian-style frustum formula involving A_1 , A_2 , and $\sqrt{A_1 A_2}$ (Olu 2020). Heydari and Muroi discuss solid-volume formulae in Elamite mathematics and identify correct computation of certain pyramid volumes in ancient mathematical tablets (Heydari and Muroi 2023).

The modern engineering expression

$$V = \frac{L}{6}(A_1 + 4A_m + A_2)$$

therefore combines ancient geometric insight with the later formalism of integral calculus and numerical quadrature. It is also closely connected to Simpson's rule, which is exact for quadratic functions. The homothetic irregular polygon case is a particularly clear example because the area function is exactly quadratic in the normal coordinate.

Proof by decomposition into triangular frusta

An irregular polygon can be triangulated from a fixed internal point or by any valid non-overlapping triangulation. If the end polygons are homothetic, each triangle in the first polygon corresponds to a similar triangle in the second polygon. Let the first end polygon be decomposed into m triangles with areas a_{1j} , and let the second end polygon have corresponding triangle areas a_{2j} , where $j = 1, 2, \dots, m$.

For each triangular frustum,

$$v_j = \frac{L}{3}(a_{1j} + \sqrt{a_{1j}a_{2j}} + a_{2j}).$$

Because the scale is uniform, there is a constant r such that

$$a_{2j} = r^2 a_{1j}.$$

Therefore

$$\sqrt{a_{1j}a_{2j}} = \sqrt{a_{1j}r^2 a_{1j}} = r a_{1j}.$$

The sum of all triangular-frustum volumes is

$$V = \sum_{j=1}^m v_j.$$

Substituting,

$$V = \frac{L}{3} \sum_{j=1}^m (a_{1j} + r a_{1j} + r^2 a_{1j}).$$

Thus

$$V = \frac{L}{3}(1 + r + r^2) \sum_{j=1}^m a_{1j}.$$

Since

$$\sum_{j=1}^m a_{1j} = A_1$$

and

$$A_2 = r^2 A_1,$$

we obtain

$$V = \frac{L}{3} (A_1 + r A_1 + r^2 A_1).$$

But

$$r A_1 = \sqrt{A_1 A_2},$$

so

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

This proof shows explicitly why the formula is valid for irregular polygons: any irregular polygon can be triangulated, and uniform scaling preserves similarity of every triangle.

Practical algorithm for engineering use

Required input data

The following data are required:

1. A_1 : area of the first irregular polygon.
2. A_2 : area of the second irregular polygon.
3. L : perpendicular distance between the two parallel planes.
4. Confirmation that the end polygons are homothetic: same shape, same vertex order, uniform scale.
5. Confirmation that corresponding vertices are joined by straight generatrices.
6. Statement of datum and distance convention: normal distance or vertical projection.

Calculation procedure

Step 1: compute the equivalent linear measures

$$q_1 = \sqrt{A_1},$$

and

$$q_2 = \sqrt{A_2}.$$

Step 2: compute the middle equivalent linear measure

$$q_m = \frac{q_1 + q_2}{2}.$$

Step 3: compute the middle area

$$A_m = q_m^2.$$

Therefore

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2.$$

Step 4: compute the prismatic volume

$$V = \frac{L}{6} (A_1 + 4A_m + A_2).$$

Step 5: optionally compute the same result directly as

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

Step 6: if starting from the average end-area volume, compute

$$V_{EA} = \frac{L}{2} (A_1 + A_2),$$

then

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2,$$

and

$$V = V_{EA} - C.$$

Decision logic for practical volume calculation

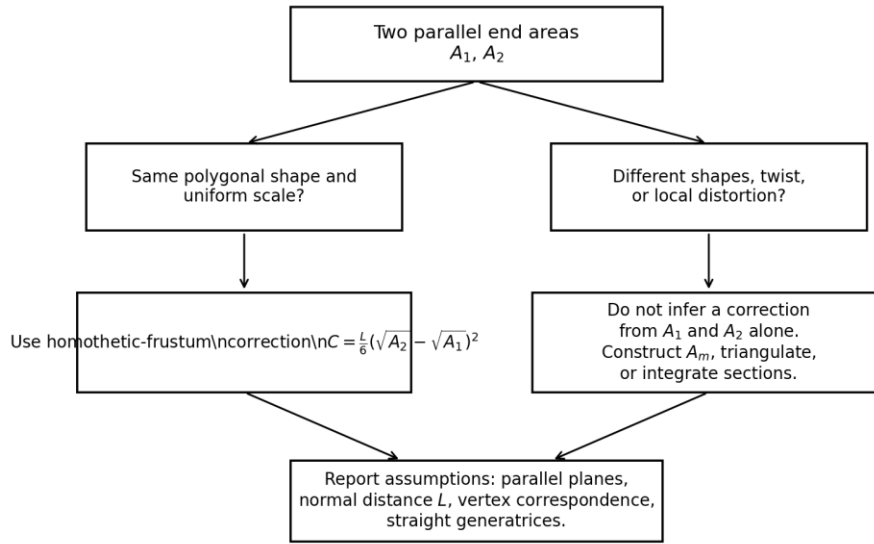


Figure 7. Decision chart for selecting the homothetic-frustum correction or a more general geometric/numerical method.

Recommended report wording

The following wording is suitable for a calculation note:

Because the end areas A_1 and A_2 are parallel and correspond to the same irregular plane shape scaled uniformly between levels, the unknown mid-section area has been obtained by averaging equivalent linear dimensions, not by averaging the areas. The adopted middle area is

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2.$$

The prismoidal volume is therefore

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

Equivalently, the average end-area volume has been reduced by

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

This is a homothetic-frustum correction. It is not the three-level road-section correction and should not be applied to non-homothetic irregular polygons without constructing or otherwise estimating the true middle area.

Limitations and validity checks

Validity checklist

Before applying the correction, the following conditions should be checked:

- The two end planes are parallel.
- L is measured normal to the planes unless vertical projection is explicitly intended.
- The two polygons have the same number of vertices.
- Vertex correspondence is known and ordered consistently.
- The second polygon is obtained from the first by uniform scaling, not by independent movement of vertices.
- There is no twist, rotation-induced warping, local distortion, or change of shape.
- The side surfaces are generated by straight lines joining corresponding vertices.

If these checks are satisfied, the correction is exact for the homothetic model.

Failure modes

The correction should not be used in the following cases:

1. The end polygons are different shapes.
2. The end polygons have different numbers of vertices.
3. One polygon has local deformation not present in the other.
4. The correspondence between vertices is ambiguous.
5. The side surfaces are curved in a way not represented by straight generatrices.
6. The planes are not parallel and no curvature/obliquity correction is made.
7. The solid is being modelled along a curved alignment without accounting for curvature.

Deakin and Hunter warn that the prismoidal formula is not universally correct for solids having curved faces and that special care is needed when the actual solid differs from the assumed prismoid (Deakin and Hunter 2009). Venkatramaiah similarly frames prismoidal and trapezoidal methods in terms of practical earthwork approximations, noting that real earth solids may not exactly be prismoids (Venkatramaiah 2011).

Comparison of methods

The principal volume methods can be compared as follows. Formulae are shown separately so that no assumption is hidden inside tabular notation.

Prism. Required data: one base area and a perpendicular height. The formula is

$$V = Bh.$$

It is exact when the cross-sectional area is constant.

Average end-area method. Required data: A_1 , A_2 , and L . The formula is

$$V_{EA} = \frac{L}{2} (A_1 + A_2).$$

It is exact when the area varies linearly between the end sections. In earthworks it is often used as a convenient approximation (Agor 2015; Venkatramaiah 2011).

Prismoidal method. Required data: A_1 , A_m , A_2 , and L . The formula is

$$V_P = \frac{L}{6} (A_1 + 4A_m + A_2).$$

It is exact for prismoid conditions and for quadratic area variation. It is Simpson's one-panel rule for volume (Deakin and Hunter 2009; Agor 2015).

Homothetic-frustum method. Required data: A_1 , A_2 , L , and confirmation that the two end polygons are similar by uniform scaling. The formula is

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

It is exact for similar irregular polygons connected by straight corresponding generatrices.

Finite element or TIN method. Required data: vertices, surfaces, and triangulation. The generic summation form is

$$V = \sum_j v_j.$$

It is preferred when the end shapes are non-similar or when the intermediate surface must be controlled explicitly.

Numerical integration. Required data: a sequence of areas at known stations. The governing expression is

$$V = \int_{x_0}^{x_n} A(x) dx.$$

The integral may be approximated by the trapezoidal rule, Simpson's rule, or a finite-element summation depending on the data spacing and area variation (Malvic et al. 2014).

Discussion

The central misconception in practical use is the assumption that the middle area of a prismoid is the average of the two end areas. That assumption is valid only in particular geometric cases. The supplied sources repeatedly show the correct distinction: dimensions are averaged for the middle section, not areas (Venkatramaiah 2011; Volume Calculations

Manual 2024). When the end figures are similar, averaging dimensions is equivalent to averaging square roots of areas.

The homothetic irregular-polygon case is especially useful because it provides a rigorous correction without requiring the physical middle polygon. It avoids both extremes: it is more accurate than simply using the average end-area formula, but it does not require the extra fieldwork or CAD construction of the actual mid-section. Its validity rests entirely on the similarity assumption.

The formula is also robust for irregular shapes. Irregularity alone is not a problem. A highly irregular polygon behaves exactly like a square, triangle, or circle under uniform scaling: all areas scale with the square of the linear scale. Therefore the formula is not limited to regular solids. It is limited to homothetic solids.

The broader practical lesson is that no prismoidal correction should be transferred from one geometry to another merely because the symbols look similar. A road-section correction belongs to its road-section model. A pyramid-frustum correction belongs to homothetic sections. A Simpson-rule volume belongs to an assumed or measured quadratic area function. A finite-element model belongs to the triangulation and interpolation rules used to define it.

Conclusions

The standard prismoidal formula is

$$V_P = \frac{L}{6}(A_1 + 4A_m + A_2).$$

The average end-area formula is

$$V_{EA} = \frac{L}{2}(A_1 + A_2).$$

The general positive correction to subtract from the average end-area volume is

$$C = \frac{L}{3}(A_1 + A_2 - 2A_m).$$

For irregular polygons of the same plane shape in parallel planes, differing only by uniform linear scale, the correct middle area is

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2.$$

The resulting exact volume is

$$V = \frac{L}{3}(A_1 + \sqrt{A_1 A_2} + A_2).$$

The corresponding correction to the average end-area method is

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2.$$

Thus

$$V = V_{EA} - C.$$

This is the correction to apply when the end areas are irregular but homothetic. It is not the usual three-level road-section correction. It must not be used for arbitrary irregular polygons that are not uniformly scaled versions of one another. In those cases, the middle area must be constructed, surveyed, triangulated, or estimated by a defensible numerical integration or finite-element method.

Illustrated catalogue of classical solids and worked calculations

The article above develops the theory of prismoidal volumes and the special homothetic correction for similar irregular polygons. For practical use, it is also valuable to visualise the most common solids that appear in engineering geometry and to include fully worked numerical examples. The solids collected below include exact prismoids and prismatoids in the strict geometric sense, as well as classical curved solids such as cylinders, cones, and spheres. Although some of the curved solids are not polyhedral prismoids, they are closely related to prismoidal reasoning because their cross-sectional area laws are either constant, linear, or quadratic; therefore the same area-based logic explains why the classical formulas arise and, in several cases, why Simpson/prismoidal integration is exact (Deakin and Hunter 2009; Volume Calculations Manual 2024; Frustum and Prismatoid Module n.d.; Olu 2020; Heydari and Muroi 2023).

Classical Solids and Prismoidal Relatives

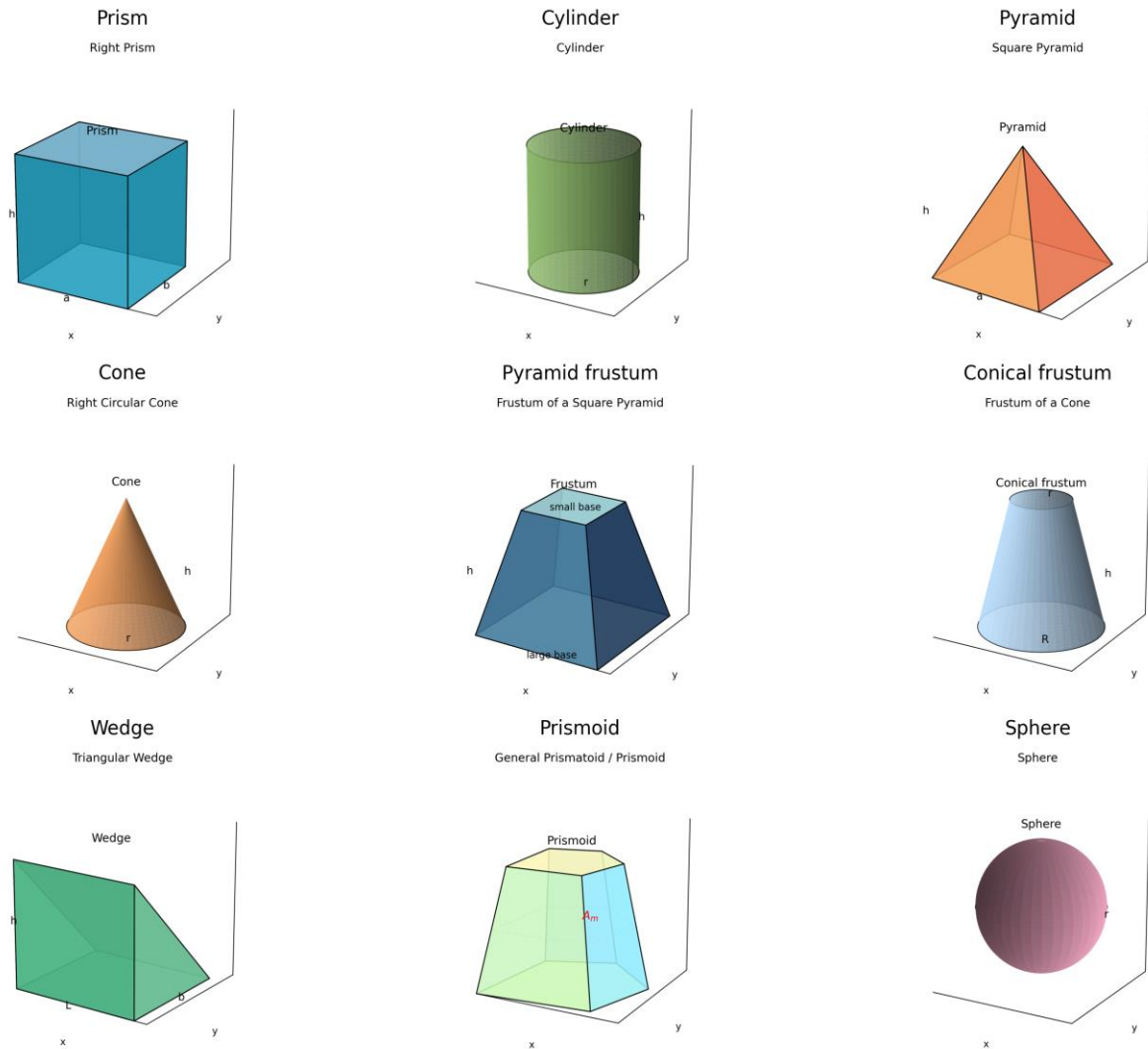


Figure 18. Overview plate of classical solids and prismoidal relatives discussed in this article: right prism, cylinder, pyramid, cone, pyramid frustum, conical frustum, wedge, general prismoid, and sphere.

Right prism

A right prism has two parallel congruent polygonal bases and lateral faces perpendicular to the bases. Its volume is the product of one base area and the perpendicular height (Volume Calculations Manual 2024; Bonjorno, Giovanni Junior, and Camara de Sousa 2020):

$$V = Bh.$$

If the prism is rectangular with side lengths a and b , then $B = ab$ and

$$V = abh.$$

Right Prism

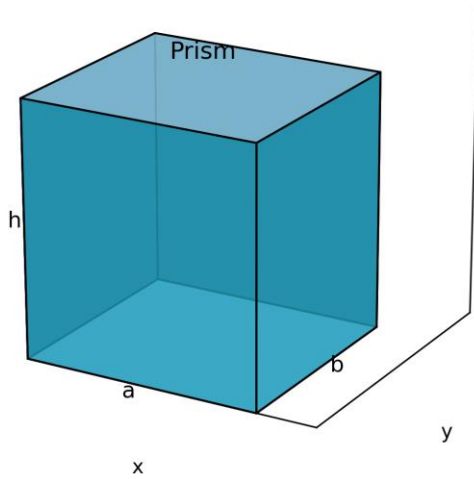


Figure 19. Right prism.

Worked example. Let $a = 4$ m, $b = 3$ m, and $h = 5$ m. Then

$$B = ab = (4)(3) = 12 \text{ m}^2,$$

and therefore

$$V = Bh = (12)(5) = 60 \text{ m}^3.$$

A prism is a limiting case of the prismoidal formula because the end and middle areas are all equal: $A_1 = A_m = A_2 = B$. Hence

$$V = \frac{L}{6}(B + 4B + B) = BL.$$

Cylinder

A cylinder is the curved analogue of a prism. If r is the radius of the circular base and h is the height, then the classical formula is (Volume Calculations Manual 2024)

$$V = \pi r^2 h.$$

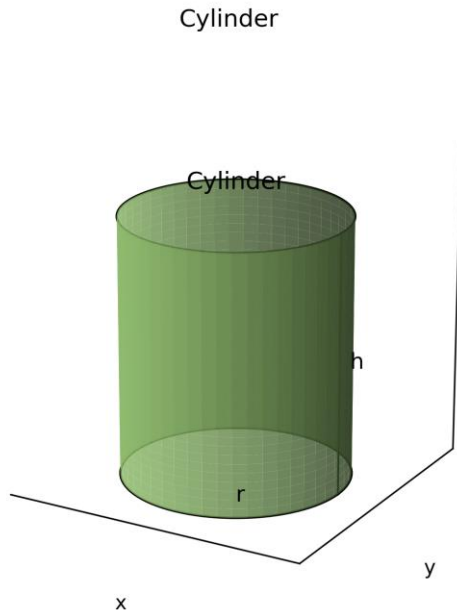


Figure 20. Cylinder.

Worked example. Let $r = 3$ m and $h = 10$ m. Then

$$V = \pi r^2 h = \pi(3)^2(10) = 90\pi \text{ m}^3 \approx 282.743 \text{ m}^3.$$

The cylinder is not a polyhedral prismoid, but because the cross-sectional area is constant, the average end-area formula and the prismoidal formula are both exact.

Pyramid

A pyramid has polygonal base area B and apex height h . Its volume is (Volume Calculations Manual 2024; Heydari and Muroi 2023)

$$V = \frac{1}{3}Bh.$$

Square Pyramid

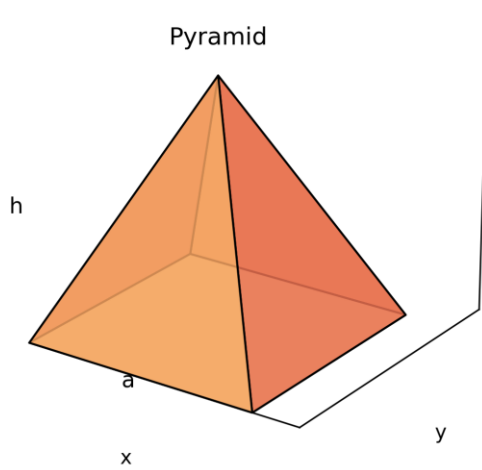


Figure 21. Square pyramid.

Worked example. Let the base be a square of side $a = 6$ m and let the height be $h = 9$ m. Then

$$B = a^2 = (6)^2 = 36 \text{ m}^2,$$

and therefore

$$V = \frac{1}{3}Bh = \frac{1}{3}(36)(9) = 108 \text{ m}^3.$$

The same result follows from the prismoidal formula by taking one end area to be zero and the middle area to be one quarter of the base area:

$$A_1 = 0, \quad A_2 = B, \quad A_m = \frac{B}{4},$$

so that

$$V = \frac{h}{6} \left(0 + 4 \cdot \frac{B}{4} + B \right) = \frac{1}{3}Bh.$$

Cone

A right circular cone has base radius r and height h . Its classical volume is

$$V = \frac{1}{3}\pi r^2 h.$$

Right Circular Cone

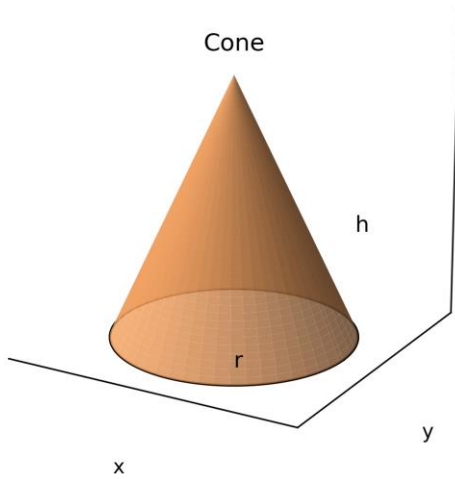


Figure 22. Right circular cone.

Worked example. Let $r = 3$ m and $h = 8$ m. Then

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(3)^2(8) = 24\pi \text{ m}^3 \approx 75.398 \text{ m}^3.$$

The cone is also recovered exactly by the prismoidal formula because the cross-sectional area varies quadratically with distance from the apex. With $A_1 = 0$, $A_2 = \pi r^2$, and $A_m = \pi(r/2)^2 = \pi r^2/4$, one obtains

$$V = \frac{h}{6} \left(0 + 4 \cdot \frac{\pi r^2}{4} + \pi r^2 \right) = \frac{1}{3}\pi r^2 h.$$

Frustum of a pyramid

For a pyramidal frustum with parallel similar end faces of areas A_1 and A_2 , and height h , the exact formula is (Olu 2020; Frustum and Prismatoid Module n.d.)

$$V = \frac{h}{3} (A_1 + \sqrt{A_1 A_2} + A_2).$$

Frustum of a Square Pyramid

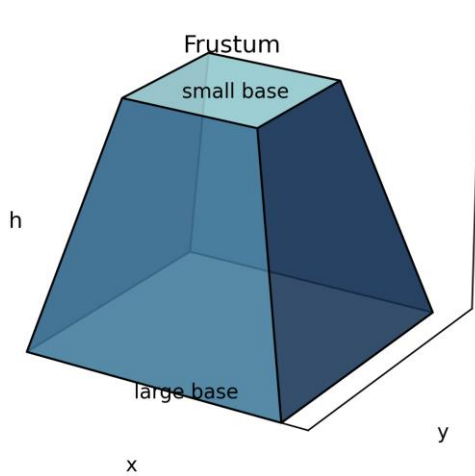


Figure 23. Frustum of a square pyramid.

Worked example. Let $A_1 = 100 \text{ m}^2$, $A_2 = 36 \text{ m}^2$, and $h = 12 \text{ m}$. Then

$$\sqrt{A_1 A_2} = \sqrt{(100)(36)} = 60 \text{ m}^2,$$

and therefore

$$V = \frac{12}{3} (100 + 60 + 36) = 4(196) = 784 \text{ m}^3.$$

The corresponding middle area is

$$A_m = \left(\frac{\sqrt{100} + \sqrt{36}}{2} \right)^2 = \left(\frac{10 + 6}{2} \right)^2 = 64 \text{ m}^2,$$

and substitution in the prismoidal formula confirms the same result:

$$V = \frac{12}{6} (100 + 4 \cdot 64 + 36) = 2(392) = 784 \text{ m}^3.$$

Frustum of a cone

For a conical frustum with bottom radius R , top radius r , and height h , the usual formula is

$$V = \frac{\pi h}{3} (R^2 + Rr + r^2).$$

It is equivalent to the area formula

$$V = \frac{h}{3} (A_1 + \sqrt{A_1 A_2} + A_2),$$

with $A_1 = \pi R^2$ and $A_2 = \pi r^2$ (Olu 2020).

Frustum of a Cone

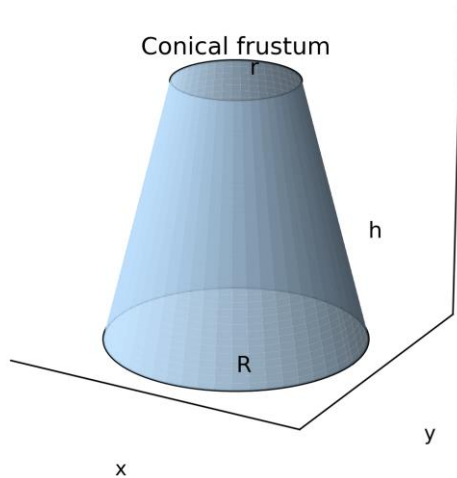


Figure 24. Frustum of a cone.

Worked example. Let $R = 5$ m, $r = 2$ m, and $h = 10$ m. Then

$$V = \frac{\pi h}{3} (R^2 + Rr + r^2) = \frac{10\pi}{3} (25 + 10 + 4) = 130\pi \text{ m}^3 \approx 408.407 \text{ m}^3.$$

Using areas gives the same answer:

$$A_1 = 25\pi, \quad A_2 = 4\pi, \quad \sqrt{A_1 A_2} = 10\pi,$$

hence

$$V = \frac{10}{3} (25\pi + 10\pi + 4\pi) = 130\pi \text{ m}^3.$$

Triangular wedge

A wedge may be viewed as a prismatoid generated when one dimension of a prism tapers to zero along an edge. For a triangular wedge with triangular section of base b and height h , extruded through length L , the volume is

$$V = \frac{1}{2} b h L.$$

Triangular Wedge

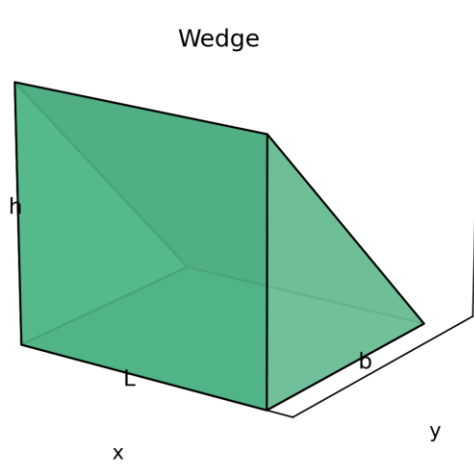


Figure 25. Triangular wedge.

Worked example. Let $b = 6$ m, $h = 4$ m, and $L = 10$ m. Then

$$V = \frac{1}{2}(6)(4)(10) = 120 \text{ m}^3.$$

Wedges are important because they appear repeatedly in decompositions of general prismoids in surveying and geometric proofs (Deakin and Hunter 2009).

General prismoid

The general prismoid is the central engineering solid of this article. It has end areas A_1 and A_2 , middle area A_m , and spacing L . Its volume is

$$V = \frac{L}{6}(A_1 + 4A_m + A_2).$$

General Prismatoid / Prismoid

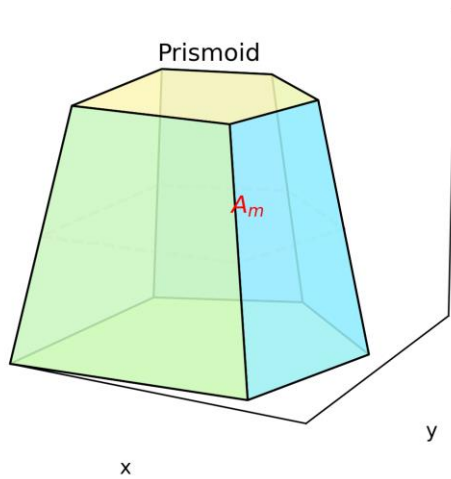


Figure 26. General prismatoid / prismoid with a middle section A_m .

Worked example. Let

$$A_1 = 20 \text{ m}^2, \quad A_m = 26 \text{ m}^2, \quad A_2 = 40 \text{ m}^2, \quad L = 15 \text{ m}.$$

Then

$$V = \frac{15}{6} (20 + 4 \cdot 26 + 40) = 2.5(164) = 410 \text{ m}^3.$$

The average end-area formula would give

$$V_{EA} = \frac{15}{2} (20 + 40) = 450 \text{ m}^3,$$

so the positive correction to subtract is

$$C = V_{EA} - V = 450 - 410 = 40 \text{ m}^3.$$

This example makes explicit the practical purpose of the prismoidal correction.

Sphere

A sphere of radius r has volume

$$V = \frac{4}{3} \pi r^3.$$

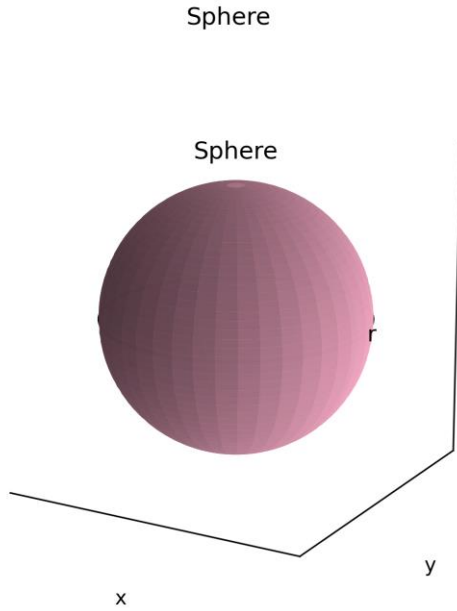


Figure 27. Sphere.

Worked example. Let $r = 4$ m. Then

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(4)^3 = \frac{256\pi}{3} \text{ m}^3 \approx 268.083 \text{ m}^3.$$

The sphere is not a prismoid in the strict polyhedral sense, but it provides a beautiful illustration of why Simpson/prismoidal reasoning is so powerful. Taking the full diameter as the integration interval, the end sections are points with zero area and the middle section is a great circle of area πr^2 . Therefore,

$$A_1 = 0, \quad A_m = \pi r^2, \quad A_2 = 0, \quad L = 2r,$$

and the prismoidal formula gives exactly

$$V = \frac{2r}{6} (0 + 4\pi r^2 + 0) = \frac{4}{3}\pi r^3.$$

This exactness follows because the sectional area function of a sphere,

$$A(z) = \pi(r^2 - z^2),$$

is quadratic in z .

Worked example for the homothetic irregular-polygon correction

Because the user's practical problem concerns irregular polygons, it is useful to add one more fully numerical example directly in the form of the special correction derived in this article.

Assume that two similar irregular polygonal end sections have areas

$$A_1 = 25 \text{ m}^2, \quad A_2 = 64 \text{ m}^2,$$

and lie in parallel planes separated by

$$L = 9 \text{ m}.$$

Then the correct middle area is

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2 = \left(\frac{5 + 8}{2} \right)^2 = 42.25 \text{ m}^2.$$

Hence the exact volume is

$$V = \frac{L}{6} (A_1 + 4A_m + A_2) = \frac{9}{6} (25 + 4 \cdot 42.25 + 64) = 387 \text{ m}^3.$$

Using the homothetic-frustum form gives the same answer immediately:

$$V = \frac{L}{3} (A_1 + \sqrt{A_1 A_2} + A_2) = \frac{9}{3} (25 + 40 + 64) = 387 \text{ m}^3.$$

If the average end-area formula had been used first, the result would have been

$$V_{EA} = \frac{9}{2} (25 + 64) = 400.5 \text{ m}^3.$$

Therefore the positive correction to subtract is

$$C = V_{EA} - V = 400.5 - 387 = 13.5 \text{ m}^3,$$

which also agrees with the closed-form correction:

$$C = \frac{L}{6} (\sqrt{A_2} - \sqrt{A_1})^2 = \frac{9}{6} (8 - 5)^2 = 13.5 \text{ m}^3.$$

This worked example is the direct operational answer to the original irregular-polygon problem: if the two end polygons are homothetic, the correction is obtained from the square roots of the end areas, not from the arithmetic mean of the areas.

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